THE CHINESE UNIVERSITY OF HONG KONG
Department of Statistics

STAT3003: Survey Methods
7. Using Auxiliary Data - Exercises

1. Ratio Estimation in simple random sampling

For each observation unit, we have two measurements: y and x. Variable y is the variable of interest. Variable x is an **auxiliary variable** or **subsidiary variable**, which is used to improve our estimate of population quantities of y. We use y_i and x_i to denote respectively the y value and x value of the i^{th} unit in the population.

Estimator of ratio ($R = \tau_y/\tau_x$)

This section provides the formula for the ratio estimator $\hat{R}$ and two forms for its estimated variance.

$$\text{Estimator } \hat{R} = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} x_i} = \frac{\bar{y}}{\bar{x}}$$

The estimated variance of $\hat{R}$ can be expressed as:

$$\widehat{Var}(\hat{R}) = \frac{1}{\mu_x^2}\left(1 - \frac{n}{N}\right)\frac{1}{n}\hat{\sigma}_r^2$$

Or, alternatively:

$$\widehat{Var}(\hat{R}) = \left(1 - \frac{n}{N}\right)\frac{1}{n}\left(\frac{1}{\mu_x^2}\right)\left(\hat{\sigma}_y^2 + R^2\hat{\sigma}_x^2 - 2R\hat{\rho}\hat{\sigma}_x\hat{\sigma}_y\right)$$

where $\hat{\sigma}_r^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \hat{R}X_i)^2$, $\hat{\rho} = \frac{\hat{\sigma}_{xy}}{\hat{\sigma}_x\hat{\sigma}_y}$, $\hat{\sigma}_x$ and $\hat{\sigma}_y$ are the sample standard deviations of x and y, and $\hat{\sigma}_{xy}$ is the sample covariance.

Estimator of τ_y

The population total τ_y and its ratio estimator $\hat{\tau}_y$ are defined as follows:

Population	**Sample (Estimate)**
$\tau_y = \frac{\mu_y}{\mu_x}\tau_x = R\tau_x$	$\hat{\tau}_y = \frac{\bar{y}}{\bar{x}}\tau_x = \hat{R}\tau_x$

So, we have the ratio estimator of the total, denoted $\hat{\tau}_r$, and its estimated variance:

$$\hat{\tau}_r = \hat{R}\tau_x \quad \text{and} \quad \hat{V}(\hat{\tau}_r) = \tau_x^2\widehat{Var}(\hat{R}) = N^2\mu_x^2\widehat{Var}(\hat{R}) = N^2\left(1 - \frac{n}{N}\right)\frac{1}{n}\hat{\sigma}_r^2$$

Estimator of μ_y

Based on the proportionality assumption between the means, we can define the ratio estimator for the population mean μ_y. Since $\mu_y : \mu_x \approx \bar{y} : \bar{x}$, we have:

$$\hat{\mu}_r = \hat{R}\mu_x \quad \text{and} \quad \widehat{Var}(\hat{\mu}_r) = \mu_x^2\widehat{Var}(\hat{R}) = \left(1 - \frac{n}{N}\right)\frac{1}{n}\hat{\sigma}_r^2$$

Note for "better" ratio estimator:

This section provides a condition under which the ratio estimator is preferable to the simple expansion estimator.

- In a simple random sample setting, $\hat{R}\tau_x$ can have a smaller variance than $N\bar{y}$ if the correlation between x and y is **strongly positive**.
- Let $\hat{c}_{v,x} := \frac{\hat{\sigma}_x}{\bar{x}}$ be the coefficient of variation. If the sample correlation coefficient $\hat{\rho}$ satisfies the condition $\hat{\rho} \gg \frac{1}{2}\frac{\hat{c}_{v,x}}{\hat{c}_{v,y}}$, then it follows that $\hat{\sigma}_r^2 \ll \hat{\sigma}_y^2$, and thus ratio estimation is a good choice.

2. Ratio Estimation in Cluster Sampling

Ratio estimator is a good choice for estimating if the cluster total ($\sum Y_i$) and the cluster population size ($\sum M_i$) are highly correlated. In other words, Y_i and M_i are highly correlated.

2.1 One-Stage Cluster Sampling

Estimator of τ_r

The ratio estimator for the population total, τ_r, and its estimated variance are given by:

$$\hat{\tau}_r = \frac{\bar{Y}}{\bar{M}}M = \hat{R}M$$

$$\widehat{Var}(\hat{\tau}_r) = N(N-n)\frac{1}{n}\hat{\sigma}_r^2$$

where $M = \sum_{j=1}^{N} M_j$, $\bar{M} = \frac{1}{n}\sum_{i=1}^{n} M_i$ and $\hat{\sigma}_r^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \hat{R}M_i)^2$.

Estimator of p_r

The ratio estimator for a population proportion, p_r, and its estimated variance are:

$$\hat{p}_r = \frac{\sum_{i=1}^{n} A_i}{\sum_{i=1}^{n} M_i}$$

$$\widehat{Var}(\hat{p}_r) = \frac{N-n}{N}\frac{N^2}{M^2}\frac{\hat{\sigma}_r^2}{n}$$

Where $A_i = \#$ of people in the cluster having a certain characteristic and $\hat{\sigma}_r^2 = \frac{1}{n-1}\sum_{i=1}^{n}(A_i - \hat{p}_r M_i)^2$.

2.2 Two-Stage Cluster Sampling

The ratio estimator for the population mean, μ_r, and its estimated variance are:

$$\hat{\mu}_r = \frac{\sum_{i=1}^{n} M_i\bar{Y}_i}{\sum_{i=1}^{n} M_i} = \frac{\sum_{i=1}^{n} \hat{Y}_i}{\sum_{i=1}^{n} M_i}$$

$$\widehat{Var}(\hat{\mu}_r) = \frac{1}{M^2}N(N-n)\frac{1}{n}\hat{\sigma}_r^2 + \frac{1}{M^2}\frac{N}{n}\sum_{i=1}^{n} M_i(M_i - m_i)\frac{1}{m_i}\hat{\sigma}_i^2$$

Where $\hat{\sigma}_r^2 = \frac{1}{n-1}\sum_{i=1}^{n}(\hat{Y}_i - \hat{\mu}_r M_i)^2$ and $\hat{\sigma}_i^2$ is the variance inside cluster i.

Note: If M_i and $\hat{Y}_i$ are highly correlated and $\hat{\rho} \gg \frac{1}{2}\frac{\hat{c}_{v,M}}{\hat{c}_{v,\hat{Y}}}$, then ratio estimation is a good choice.

3. Regression Estimation

Regression estimator can be considered as the general form of ratio estimator (with intercept adjustment). It is used after observing the "linear pattern" shown in the scatterplot.

Simple linear regression: $Y_i = a + bX_i + \epsilon_i$, where $\epsilon_i \sim N(0, \sigma^2)$

Least-square method (minimize $\sum_{i=1}^n \epsilon_i^2$) is used to estimate a and b, where

$$\hat{a} = \bar{y} - \hat{b}\bar{x}$$

$$\hat{b} = \frac{S_{XY}}{S_{XX}} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n x_i y_i - \frac{1}{n}\sum_{i=1}^n x_i \sum_{i=1}^n y_i}{\sum_{i=1}^n x_i^2 - \frac{1}{n}(\sum_{i=1}^n x_i)^2}$$

The predicted values of y_i can be calculated from $\hat{y}_i = \hat{a} + \hat{b}x_i = \bar{y} + \hat{b}(x_i - \bar{x})$.

Thus the regression estimator for a population mean μ_y is:

$$\hat{\mu}_L = a + b\mu_x = \bar{y} + \hat{b}(\mu_x - \bar{x}) \tag{1}$$

And the estimated variance of the population mean is:

$$\begin{aligned}\widehat{Var}(\hat{\mu}_L) &= \frac{N-n}{Nn}\frac{1}{n-2}\sum_{i=1}^n (Y_i - a - bx_i)^2 \\ &= \frac{N-n}{Nn}\frac{1}{n-2}\sum_{i=1}^n (Y_i - \bar{y} - b(\mu_x - \bar{x}))^2 \\ &\approx \frac{N-n}{Nn}\sigma_y^2(1-\hat{\rho}^2) \text{ Derive it when you use it}\end{aligned} \tag{2}$$

An approximate $(1-\alpha)100\%$ confidence interval for μ_y is given by:

$$\hat{\mu}_L \pm t_{n-2,1-\frac{\alpha}{2}}\sqrt{\widehat{Var}(\hat{\mu}_L)}$$

Note: The degree of freedom is $n-2$ here.

Exercise

1. (Adapted from Final 2024 Q5) For a population of size 1,000,000, a simple random sample of 1000 units is selected. Two variables Y and X are measured from each unit in the sample. Suppose we know the population mean value of Y is 15. The following summary statistics from the sample are provided:

	X	Y
Sample mean	10	14
Sample variance	20	25
Sample covariance	15	

5.1) (10 points). Provide an unbiased estimate of the population mean of X. Remember to quantify the accuracy of your estimate.

5.2) (10 points). Provide a ratio estimate of the population mean of X. Remember to quantify the accuracy of your estimate.

5.3) (10 points). Provide a regression estimate of the population mean of X. Remember to quantify the accuracy of your estimate. (Hints, you may use equation 2 to do the estimation of variance, derive it when you use it!)

2. (Adapted from Exercise 6.9 in Scheaffer et al. (2012)) A forest resource manager is interested in estimating the number of dead fir trees in a 300-acre area of heavy infestation. Using an aerial photo, she divides the area into 200 1.5-acre plots. Let x denote the photo count of dead firs and y the actual ground count for a SRS of $n = 10$ plots. The total number of dead fir trees obtained from the photo count is $\tau_x = 4200$. Use the sample data in Table 1 to estimate τ_y, the total number of dead firs in the 300-acre area. Provide a 95% confidence interval.

Plot	Photo count	Ground count	Plot	Photo count	Ground count
1	12	18	6	30	36
2	30	42	7	12	14
3	24	24	8	6	10
4	24	36	9	36	48
5	18	24	10	42	54

Table 1: Data from the Dead Trees Survey

3. Show that under SRS with regression estimation, the following equations apply for estimating τ:

$$\hat{\tau}_L = aN + b\tau_x$$
$$\widehat{Var}(\hat{\tau}_L) = N(N-n)\frac{1}{n}\frac{1}{n-2}\sum_{i=1}^{n}(Y_i - a - bx_i)^2$$

where a and b are the estimated intercept and slope coefficient from the linear regression $y = \alpha + \beta x$.

4. (Adapted from Exercise 6.23 in Scheaffer et al. (2012)) National income from manufacturing industries is to be estimated for 1989 from a sample of 6 of the 19 industry categories that reported figures for that year. Incomes from all 19 industries are known for 1980 and the total is \$674 billion. From the data provided, estimate the total national income from manufacturing in 1989, with a 95% confidence interval. All figures are in billions of constant (1982) dollars.

Industry	1980	1989
Lumber and wood products	21	26
Electric and electronic equipment	63	91
Motor vehicles and equipment	91	47
Food and kindred products	60	70
Textile mill products	70	70
Chemicals and allied products	50	50

Table 2: Data from US Bureau of the Census, Statistical Abstract of the Unites States, 1993-4, Washington D.C., 199

(a) Find a ratio estimator of the 1989 total income and a 95% confidence interval.

(b) Find a regression estimator the 1989 total income and a 95% confidence interval.

5. (Adapted from Exercises 9.2 and 9.3 from Scheaffer at al. (2012)) A gardener wants to estimate the average height of seedlings in a large field that is divided into 50 plots that vary slightly in size. He believes the heights are fairly constant throughout each plot but may vary considerably from plot to plot. Therefore, he decides to sample 10% of the trees within each of 10 plots using a two-stage cluster sample. The gardener knows there are about 2500 seedlings in the field. The data are given in Table 3. Provide a 95% confidence interval for the mean seedling height in the field.

Plot #	# Seedlings	# Seedling Sampled	Heights (in.)
1	52	5	12, 11, 12, 10, 13
2	56	6	10, 9, 7, 9, 8, 10
3	60	6	6, 5, 7, 5, 6, 4
4	46	5	7, 8, 7, 7, 6
5	49	5	10, 11, 13, 12, 12
6	51	5	14, 15, 13, 12, 13
7	50	5	6, 7, 6, 8, 7
8	61	6	9, 10, 8, 9, 9, 10
9	60	6	7, 10, 8, 9, 9, 10
10	45	6	12, 11, 12, 13, 12, 12

Table 3: Seedling Study Data

- Do The excersize again from the tutorial Clustering sampling but this time use a ratio estimator. Compare your answer to that of the unbiased estimator.

6. Show that under two-stage cluster sampling with ratio estimation, the following equations apply for estimating τ and p:

$$\hat{\tau}_r = M\frac{\sum_{i=1}^n \hat{Y}_i}{\sum_{i=1}^n M_i}$$

$$\widehat{Var}(\hat{\tau}_r) = N(N-n)\frac{1}{n}\hat{\sigma}_r^2 + \frac{N}{n}\sum_{i=1}^n M_i(M_i - m_i)\frac{1}{m_i}\hat{\sigma}_i^2$$

where $\hat{\sigma}_r^2 = \frac{1}{n-1}\sum_{i=1}^n(\hat{Y}_i - \hat{\mu}_r M_i)^2$ and $\hat{Y}_i$ is the estimated cluster i total;

$$\hat{p}_r = \frac{\sum_{i=1}^n M_i\hat{p}_i}{\sum_{i=1}^n M_i}$$

$$\widehat{Var}(\hat{p}_r) = N(N-n)\frac{1}{nM^2}\hat{\sigma}_r^2 + \frac{N}{nM^2}\sum_{i=1}^n M_i(M_i - m_i)\frac{1}{m_i}\hat{\sigma}_i^2$$

where $\hat{\sigma}_r^2 = \frac{1}{n-1}\sum_{i=1}^{n}(M_i\hat{p}_i - \hat{p}_r M_i)^2$ and $\hat{\sigma}_i^2 = \frac{m_i}{m_i-1}\hat{p}_i(1-\hat{p}_i)$.

7. (Adapted from Exercise 9.7 in Scheaffer et al. (2012)) A city zoning commission wants to estimate the proportion of property owners in a certain section of a city who favor a proposed zoning charge. The section is divided into seven distinct residential areas, each containing similar residents. Because the results must be obtained in a short period of time, two-stage cluster sampling is used. Three of the seven areas are selected at random, and 20% of the property owners in each area selected are sampled. The figure of 20% seems reasonable because the people living within each area seem to be in the same socioeconomic class and hence tend to hold similar opinions on the zoning question. The results are given in Table 4. Estimate the proportion of property owners who favor the proposed zoning change provide a 95% confidence interval.

Area	# Property Owners	# Property Owners Sampled	# in Favor of Change
1	46	9	1
2	67	13	2
3	93	20	2

Table 4: Data from the Zoning Survey

THE END